

Nonlinear autonomous differential systems in $\mathbb{R}^m$.

The dynamical system associated.

$$(1) \dot{x} = f(x)$$

where $f: \mathbb{R}^m \rightarrow \mathbb{R}^m$, of class C^1 .

$$x = \begin{pmatrix} x_1 \\ \vdots \\ x_m \end{pmatrix} \quad f = \begin{pmatrix} f_1 \\ \vdots \\ f_m \end{pmatrix}$$

$$(1) \Leftrightarrow \begin{cases} \dot{x}_1 = f_1(x_1, \dots, x_m) \\ \dot{x}_2 = f_2(x_1, \dots, x_m) \end{cases}$$

$\dot{x} = f(x)$; $\varphi(t, \eta^*) = \eta^*$.
 $\lim_{t \rightarrow \infty} \varphi(t, \eta) = \eta^* \Rightarrow \eta^*$ equil.
 $A_{\eta^*} = \{ \eta \in \mathbb{R}^m : \lim_{t \rightarrow \infty} \varphi(t, \eta) = \eta^* \}$
 $\hookrightarrow$ basin of a ; η^* attractor.
 $A_{\eta^*}^+ = \{ \eta \in \mathbb{R}^m : \lim_{t \rightarrow -\infty} \varphi(t, \eta) = \eta^* \}$
 $\hookrightarrow \eta^*$ repeller.
 $\Gamma_c = \{ x \in \mathbb{R} : H(x) = c \}$ c level set.
 $H(\varphi(t, \eta)) = H(\eta)$.

$\rightarrow$ deriv?

For any $\eta \in \mathbb{R}^m$ we consider the IVP (2): $\dot{x} = f(x)$, $x(0) = \eta$

The existence and uniqueness theorem.

$f \in C^1(\mathbb{R}^m, \mathbb{R}^m)$, $\eta \in \mathbb{R}^m$

Then, IVP (2) has a unique sol. denoted by $\varphi(t, \eta)$

defined for $t \in I_\eta$ (I_η is a maximal interval of def.)

$$I_\eta = (\alpha_\eta, \beta_\eta)$$

If φ is bounded in the future, then $\beta_\eta = +\infty$

If φ is bounded in the past, then $\alpha_\eta = -\infty$.

Dynamical systems vocabulary

$t=0$ is the initial time

$t>0$ is a time in the future

$t<0$ is a time in the past

$\varphi(t, \eta)$: $\eta \rightarrow$ the initial state of the system

$\varphi(t, \eta) \in \mathbb{R}^m \rightarrow$ the state of the system at time t when the system initiated at η .

The function $(t, \eta) \in \mathbb{R} \times \mathbb{R}^m \mapsto \varphi(t, \eta) \in \mathbb{R}^m \rightarrow$ the flow of the system

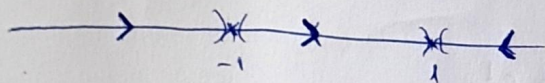
An initial state $\eta^* \in \mathbb{R}^m$ st. $\varphi(t, \eta^*) = \eta^*$, ($\forall t \in \mathbb{R}$) is said to be an equilibrium (or stationary state) of the system.

Proposition: η^* is an equilibrium of (1) $\Leftrightarrow f(\eta^*) = 0$.

(If it starts at equil, it should stay there for any moment in time)

m=1: $\dot{x} = 1 - x^2 \Rightarrow$ equil. points: $-1, 1$. $\{1\}, (1, \infty) \rightarrow ??!$

$\Rightarrow$ the orbits are: $(-\infty, -1), \{-1\}, (1, \infty)$



$1 \rightarrow$ attractor (equil. point.)
 $-1 \rightarrow$ repeller (equil. point.)

Def. of orbit: For $\eta \in \mathbb{R}^m$, the orbit $\gamma_\eta = \{ \varphi(t, \eta) : t \in I_\eta \}$ ($\sim$ Traj)

Proposition: η^* is an equil. point of (1) $\Leftrightarrow \gamma_{\eta^*} = \{\eta^*\}$ (unique)

Proposition: Let $\eta^*, \eta \in \mathbb{R}^m$ s.t. $\lim_{t \rightarrow \infty} \varphi(t, \eta) = \eta^*$

$\Rightarrow \eta^*$ is an equilibrium point.

Def: Let $\eta^* \in \mathbb{R}^m$ be an equilibrium point.

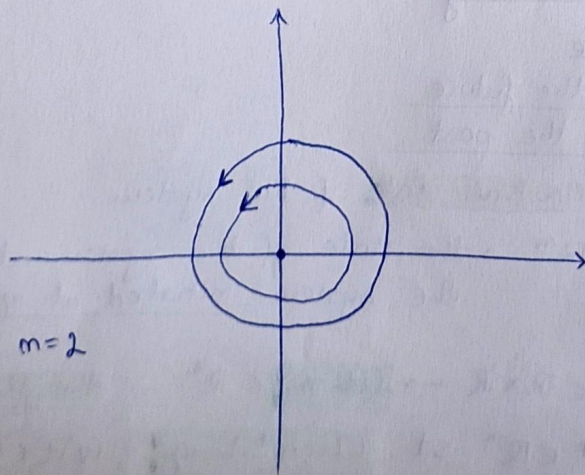
$\rightarrow$ We say that η^* is an attractor when:

the set $A_{\eta^*} = \{ \eta \in \mathbb{R}^m : \lim_{t \rightarrow \infty} \varphi(t, \eta) = \eta^* \}$ contains an open neighbourhood of η^* . In this case, A_{η^*} is said to be the basin of attraction of η^* .

Ex: $\dot{x} = 1 - x^2$, $\eta^* = 1$ is an equil., attractor, and $A_\eta = (-1, \infty)$

$\rightarrow$ We say that η^* is a repeller when:

the set $A_{\eta^*} = \{ \eta \in \mathbb{R}^m : \lim_{t \rightarrow -\infty} \varphi(t, \eta) = \eta^* \}$ contains an open neighbourhood of η^* .



?

We say η^* is stable when $(\forall) \varepsilon > 0, \exists \delta > 0$ s.t.

$$\| \varphi(t, \eta) - \eta^* \| < \varepsilon, \quad (\forall) t \in [0, \infty), \text{ whenever}$$

$$\| \eta - \eta^* \|_{\mathbb{R}^2} < \underline{\delta}$$

"Study the stability of..." $\rightarrow$ attractor? $\rightarrow$ repeller? } in exercises.

Def: Let $\Omega \in \mathbb{R}^m$ be open and $H: \Omega \rightarrow \mathbb{R}$ be a continuous function.

We say that H is a first integral in Ω of (1) if H is not locally constant (i.e. H is not constant on any open subset of Ω)

$$H(\varphi(t, \eta)) = H(\eta), \quad (\forall) t \text{ in an interval s.t. } \underline{\varphi(t, \eta)} \in \Omega.$$

The c-level set of H

$$\Gamma_c = \{x \in \Omega : H(x) = c\} \quad (\text{these values have to be in the same set})$$

(capital gamma)

! Remark: If $H: \Omega \rightarrow \mathbb{R}$ is a first integral of (1) then, $(\forall) \eta \in \Omega$, we have $\gamma_\eta \cap \Omega \subset \Gamma_{H(\eta)}$

Sketch in $\mathbb{R}^m$ the level curves of the following functions:

(a) $H: \mathbb{R}^2 \rightarrow \mathbb{R}, H(x, y) = x^2 + y^2$ 

(b) $H: \mathbb{R}^2 \rightarrow \mathbb{R}, H(x, y) = 2x^2 + 3y^2$

(c) $H: \mathbb{R}^2 \rightarrow \mathbb{R}, H(x, y) = xy$

(d) $H: (0, \infty) \times \mathbb{R} \rightarrow \mathbb{R}, H(x, y) = \frac{y}{x}$

(e) $H: (-\infty, 0) \times \mathbb{R} \rightarrow \mathbb{R}, H(x, y) = \frac{y}{x}$

(f) $H: \mathbb{R} \times (0, \infty) \rightarrow \mathbb{R}, H(x, y) = \frac{x}{y}$

a) $c \in \mathbb{R}$

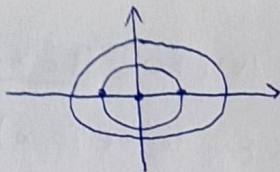
$$\Gamma_c = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = c\}$$

$c > 0 \Rightarrow \Gamma_c$ is a circle centered in $(0, 0)$ with radius $\sqrt{c}$

$c = 0 \Rightarrow \Gamma_c = \{(0, 0)\}$

$c < 0 \Rightarrow \Gamma_c = \emptyset$

b) $2x^2 + 3y^2 = 1$
 $x=0 \Rightarrow y^2 = \frac{1}{3}$
 $y=0 \Rightarrow x^2 = \frac{1}{2}$



c) $c \in \mathbb{R}$

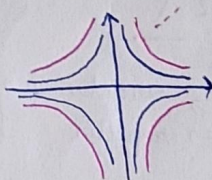
$\Gamma_c = \{(x, y) \in \mathbb{R}^2 : xy = c\}$

$c=0 \Rightarrow xy=0$

$\Gamma_0 = \{x=0\} \cup \{y=0\}$

$c=1 \Rightarrow xy=1 \Rightarrow y = \frac{1}{x}$

$f(x) = \frac{1}{x}$



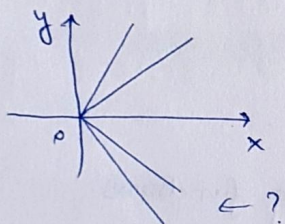
strictly decreasing (*)

x	$-\infty$	0	$+\infty$
$f(x)$	0	$+\infty$	0
$f'(x)$	---		---

(*)

$c=-1 \Rightarrow xy=-1 \Rightarrow y = -\frac{1}{x}, f(x) = -\frac{1}{x}$

d)



Ex 2: Using the def. of integral, check that $H: \mathbb{R}^2 \rightarrow \mathbb{R}, H(x, y) = xy$ is a first integral of the linear system $\begin{cases} \dot{x} = -x \\ \dot{y} = y \end{cases}$

→ Proof: we need to find the flow $\varphi(t, \eta)$. Here, $n=2$

By def, for any $\eta = \begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix}$, $\varphi(t, \eta)$ is the solution of the IVP.

$$\begin{cases} \dot{x} = -x \\ \dot{y} = y \\ x(0) = \eta_1 \\ y(0) = \eta_2 \end{cases}$$

$$\begin{cases} \dot{x} = -x \\ x(0) = \eta_1 \\ \dot{y} = y \\ y(0) = \eta_2 \end{cases}$$

$$\begin{aligned} \dot{x} + x &= 0 & x+1=0 &\Rightarrow x=-1 \\ & & \rightarrow e^{-t} &\Rightarrow x=c \cdot e^{-t} \\ x(0) &= c \cdot 1 = \eta_1 \\ & \Rightarrow x(t) = \eta_1 \cdot e^{-t} \end{aligned}$$

$$\begin{aligned} y(t) &= \eta_2 \cdot e^t \\ \varphi(t, \eta_1, \eta_2) &= \begin{pmatrix} \eta_1 \cdot e^{-t} \\ \eta_2 \cdot e^t \end{pmatrix} \quad (*) t \in \mathbb{R} \end{aligned}$$

$$H: \mathbb{R}^2 \rightarrow \mathbb{R}, H(x, y) = xy$$

H is not locally constant.

def.

We need to check: $H(\varphi(t, \eta)) = H(\eta), (\forall) t \in \mathbb{R}, (\forall) \eta \in \mathbb{R}^2$.

$$H(\eta_1 \cdot e^{-t}, \eta_2 \cdot e^t) = \eta_1 \cdot e^{-t} \cdot \eta_2 \cdot e^t = \eta_1 \cdot \eta_2 = H(\eta_1, \eta_2), (\forall) t \in \mathbb{R}, (\forall) \eta \in \mathbb{R}^2.$$

Def: Let $\Omega \subset \mathbb{R}^m$ be an open set and $H \in C^1(\Omega)$

A point $\eta^* \in \Omega$ s.t. $\nabla H(\eta^*) = 0 \in \mathbb{R}^m$ is said to be a critical point of H .

Prop: Let $\Omega \subset \mathbb{R}^m$ open, $H \in C^1(\Omega)$. If the set of critical points is finite, then H is not locally constant in Ω .

Ex 3: For all the functions H from ex 1, find the critical points and deduce that H is not locally constant.

$$c) H(x, y) = xy$$

$$\nabla H = \left(\frac{\partial H}{\partial x}, \frac{\partial H}{\partial y} \right) = (y, x)$$

$$\nabla H = (0, 0) \Leftrightarrow x = y = 0$$

$\Rightarrow$ the only critical point is $(0, 0) \Rightarrow H$ is not locally constant.

Ex 4: Using the definition, check that $H: \mathbb{R}^2 \rightarrow \mathbb{R}, H(x, y) = x^2 + y^2$ is a first integral of the linear system:

$$\begin{cases} \dot{x} = -y \\ \dot{y} = x \end{cases}$$

Proof: We need the flow of the system: φ .

$\rightarrow$ Let $\eta = \begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix}$ and IVP $\begin{cases} \dot{x} = -y \\ \dot{y} = x \end{cases}$ (Reduction method)

$$\begin{cases} \dot{x} = -y \\ \dot{y} = x \end{cases} \Rightarrow \begin{cases} y = -\dot{x} \\ \ddot{x} = -\dot{y} \\ \dot{y} = x \end{cases} \Rightarrow \ddot{x} = -x \Rightarrow \ddot{x} + x = 0.$$

$$x^2 + 1 = 0$$

$$\Rightarrow x = \pm i \rightarrow \cos t, \sin t.$$

$$x = c_1 \cos t + c_2 \sin t$$

$$\dot{x} \hookrightarrow y = c_1 \sin t - c_2 \cos t$$

$$x(0) = c_1 = \eta_1$$

$$y(0) = -c_2 = \eta_2 \Rightarrow c_2 = -\eta_2$$

$$\Rightarrow \varphi(t, \eta_1, \eta_2) = \begin{pmatrix} \eta_1 \cos t - \eta_2 \sin t \\ \eta_1 \sin t + \eta_2 \cos t \end{pmatrix}$$

(5)

$$\begin{aligned}
 H(\varphi(t, \eta)) &= (\eta_1 \cdot \cos t - \eta_2 \cdot \sin t)^2 + (\eta_1 \cdot \sin t + \eta_2 \cdot \cos t)^2 \\
 &= \eta_1^2 \cdot \cos^2 t - 2\eta_1\eta_2 \cos t \sin t + \eta_2^2 \sin^2 t + \eta_1^2 \sin^2 t + 2\eta_1\eta_2 \sin t \cos t + \eta_2^2 \cos^2 t \\
 &= \eta_1^2 \cdot (\sin^2 t + \cos^2 t) + \eta_2^2 \cdot (\sin^2 t + \cos^2 t) = \eta_1^2 + \eta_2^2, \quad (\forall) t \in \mathbb{R}, (\forall) \eta \in \mathbb{R}^2.
 \end{aligned}$$

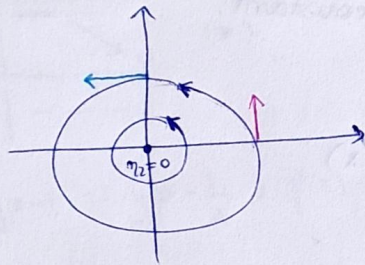
Ex 5: Sketch the phase portrait of $\begin{cases} \dot{x} = -y \\ \dot{y} = x \end{cases}$
 ("portreul fazic")

(11)

Proof: $\rightarrow$ the equil. points: $\begin{cases} -y = 0 \\ x = 0 \end{cases} \Rightarrow (0,0)$ only sol.

$\rightarrow$ We know that $H: \mathbb{R}^2 \rightarrow \mathbb{R}$, $H(x,y) = x^2 + y^2$ is a first integral.

$\Rightarrow$ the orbits are on the level curves of H .



$$\varphi(t, 1, 0) = \begin{pmatrix} \cos t \\ \sin t \end{pmatrix} \Rightarrow \text{seus trigonometric.}$$

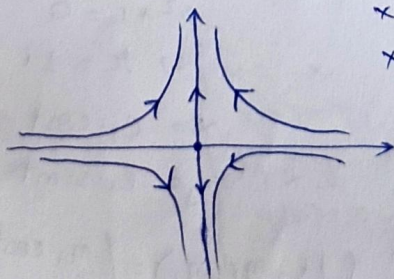
(12) $f(x,y) = \begin{pmatrix} -y \\ x \end{pmatrix}$

$$f(1,0) = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$f(0,1) = \begin{pmatrix} -1 \\ 0 \end{pmatrix}$$

Ex 6:

$$\begin{cases} \dot{x} = -x \\ \dot{y} = y \end{cases} \Rightarrow \text{equil. point: } (0,0)$$



$$\begin{aligned}
 x \text{ pos} &\Rightarrow \dot{x} \text{ neg} \Rightarrow < \\
 x \text{ neg} &\Rightarrow \dot{x} \text{ pos} \Rightarrow >
 \end{aligned}$$

(6)